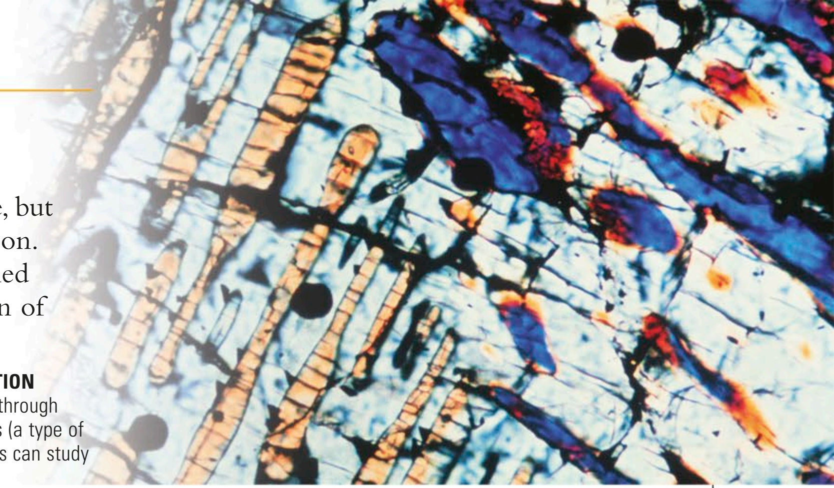
METEORITES

Meteorites are mainly pieces of asteroids that have fallen to Earth from space, but a few very rare meteorites have come from the surface of Mars and the Moon. Some meteorites are made up of the primitive material that originally formed rocky planets. These give researchers a glimpse of the conditions at the dawn of the solar system. Others are fragments of bodies that have differentiated into metallic cores and rocky surfaces, providing an indirect opportunity to study the deep interior of a rocky planet. Meteorites are named after the place where they landed.



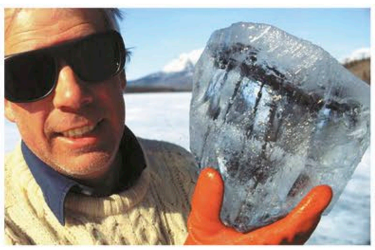
METEORITE CROSS-SECTION
By shining polarized light through thin sections of chondrites (a type of stony meteorite), scientists can study their crystalline structure.

NORTH AMERICA *north*

Tagish Lake

LOCATION British Columbia, Canada
TYPE Stony
MASS About 2.2 lb (1 kg)
DATE OF DISCOVERY 2000

Over 500 fragments of this meteorite rained down onto the frozen surface of Tagish Lake on January 18, 2000. The meteorite was dark red and rich in carbon. Analysis showed that it was extremely primitive, containing many unaltered stellar dust grains that had been part of the cloud of material that formed the Sun and the planets.



FRAGMENT ENCASED IN ICE

NORTH AMERICA *southwest*

Canyon Diablo

LOCATION Arizona
TYPE Iron
MASS 30 tons
DATE OF DISCOVERY 1891

Many pieces of this meteorite, ranging from minute fragments to chunks weighing about 1,100 lb (500 kg), have been found near Meteor Crater in Arizona. Much more is thought to be buried under one of the crater rims. If a Canyon Diablo meteorite is sawn in half and then one of the faces is polished and etched with acid, a characteristic surface pattern appears.

ACID-ETCHED, POLISHED CROSS-SECTION

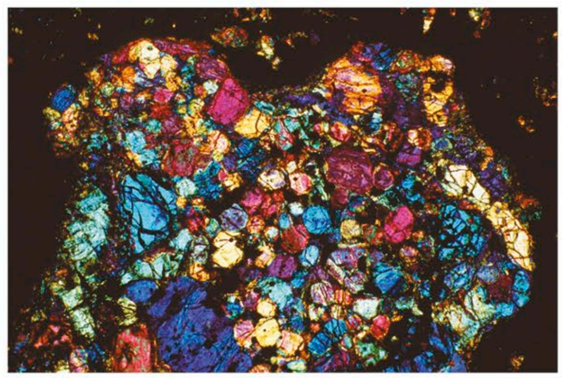
NORTH AMERICA *south*

Allende

LOCATION Chihuahua, Mexico
TYPE Stony
MASS 2 tons
DATE OF DISCOVERY 1969

On February 8, 1969, a fireball was seen streaking across the sky above Mexico. It exploded, and a shower of stones fell over an area of about 60 square miles (150 square km). Two tons of material were speedily collected and distributed among the scientific community. Allende was found to be a very rare type of primitive meteorite. Previously, only gram-sized amounts of this meteorite type were known. Because such large samples of Allende were available, destructive analysis was possible. The white calcium- and aluminum-rich crystals were separated from the surrounding rock. They were found to contain the decay products of radioactive aluminum-26, indicating that these crystals were formed in the outer shells of stars that exploded as supernovae and were subsequently incorporated into planetary material.

CHONDRULE
This thin, magnified section of an Allende meteorite shows one of many spherical, pea-sized chondrules that are locked in the stony matrix. Chondrules are droplets of silicate rock that have cooled extremely rapidly from a molten state.



EUROPE *west*

Glatton

LOCATION Cambridgeshire, UK
TYPE Stony
MASS 27 oz (767 g)
DATE OF DISCOVERY 1991

On May 5, 1991, while planting out a bed of onions just before Sunday lunch, retired English civil servant Arthur Pettifor heard a loud whining noise. Noticing one of the conifers in his hedge waving around, he got up and looked in the bottom of the hedge. He spotted a small stone that was lukewarm to the touch. If Pettifor had not been gardening, the stony meteorite would never have been found.

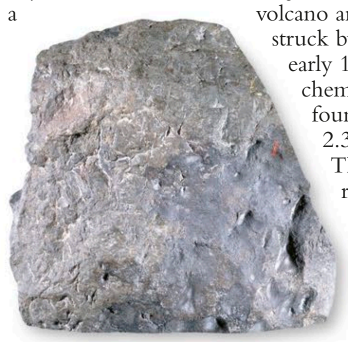
LUCKY FIND

EUROPE *west*

Ensisheim

LOCATION Alsace, France
TYPE Stony
MASS 280 lb (127 kg)
DATE OF DISCOVERY 1492

This large stone is the oldest meteorite fall that can be positively dated. It was carefully preserved by being hung from the roof of the parish church of Ensisheim, Alsace. This veneration was due to the fall being regarded by the Holy Roman Emperor Maximilian as a favorable omen for the success of his war with France and his efforts to repel Turkish invasions. Initially, Ensisheim was thought to be a "thunderstone," a rock ejected from a nearby volcano and subsequently struck by lightning. In the early 19th century, it was chemically analyzed and found to contain 2.3 percent nickel. This is very rare in rocks on Earth, and theories of an extraterrestrial origin started to proliferate.



METEORITE FRAGMENT
This highly valuable 17.6-lb (8-kg) sample of the Ensisheim meteorite is kept at the Museum of Paris, France.

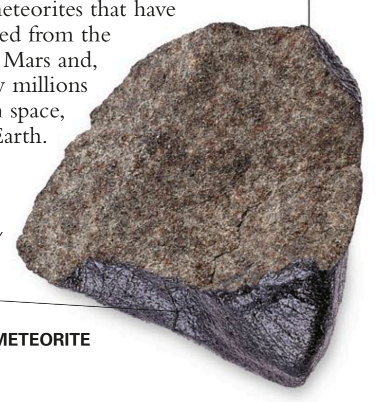
MEDIEVAL WOODCUT
The woodcut at the top of this medieval manuscript shows the meteorite falling near Ensisheim after producing a brilliant fireball in the sky on November 16, 1492.

AFRICA *north*

Nakhla

LOCATION Alexandria, Egypt
TYPE Stony
MASS 88 lb (40 kg)
DATE OF DISCOVERY 1911

On June 28, 1911, about 40 stones landed near Alexandria, the largest weighing 4 lb (1.8 kg). Nakhla is a volcanic, lavalike rock that formed 1,200 million years ago. It is one of over 16 meteorites that have been blasted from the surface of Mars and, after many millions of years in space, fallen to Earth.



MARTIAN METEORITE